

Scalability & Future Plan

Date	2 nd July 2026
Team ID	XXXXXX
Project Name	Rising waters
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	Uses historical dataset and does not fetch real-time weather data.	Prediction accuracy may reduce for current weather conditions.	High
2	No automatic SMS or email alerts for users.	Users must manually check prediction results.	Medium
3	Supports flood prediction only and does not provide detailed disaster management planning.	Limited functionality beyond prediction.	Medium

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	Supports a limited number of users on a local server	Deploy on Render/AWS and enable load balancing for multiple concurrent users.
2	Data Storage	Uses a CSV dataset.	Deploy on Render/AWS and enable load balancing for multiple concurrent users.
3	Performance	Runs predictions on a locally hosted Flask application.	Optimize the ML model and deploy on cloud infrastructure for faster response times.
4	Security	Basic input validation is implemented.	Add user authentication, HTTPS, and secure API access with proper validation.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
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Phase 2	Integrate real-time weather APIs and live rainfall data.	Next 3–6 months	Improves prediction accuracy and provides real-time analysis.
Phase 3	Add SMS, email, and mobile push notifications for flood alerts.	Next 6–9 months	Enables timely alerts and improves disaster preparedness.
Phase 4	Deploy on the cloud with user authentication, dashboard, and interactive maps.	Next 9–12 months	Improves scalability, accessibility, and overall user experience.